

Group Assignment Submission

AUTHORS

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Part A. Cleaning and Analysing Data (60 points)

1. Introduction (10 pts)

- Research Question: State the research question you are answering. Ensure it is answerable with the data you will use.

What is the relationship between the pricing of a hotel, the rating of a hotel and the family demographic of customers across bookings that are competitive and not in terms of availability?

- Method: Is your approach descriptive, causal or predictive? Explain why.

Our approach is descriptive since we want to identify the relationship of different factors whilst factoring Expedia's competitiveness. It is not predictive as we do not predict future bookings nor is it causal as we are not looking at changing any of the factors that we are analysing.

Instead, we will:

1. Clean our data and make sure it obeys Tidy data principle.
2. Construct appropriate and realistic rule-of-thumb segmentations. This will reveal to us the affects and relationships between the variables we selected
3. Finally, select optimal cluster numbers to proceed with kmeans segmentations. This will reveal to us certain groupings of Expedia which can help integrate better ranking of searches with the best options to optimise sales.

2. Data Preparation & Analysis (50 pts)

Instructions

- Show and explain your data cleaning, wrangling, and variable creation.
- Present models, visualisations, and diagnostics.
- Ensure code is reproducible and commented.

Load Libraries:

```
# load the libraries you need for your analysis here

# If you need to install a package run install.packages('package_name') once
# then load it via library(package_name)

# For rule-by-thumb segmentation
```

```
library(tidyverse) # General functions
library(dplyr) # Shaping data
library(ggokabeito) # Colour-blind friendly
library(readr) # Reading files
library(knitr) # Better looking tables

# For kmeans segmentation
library(recipes) # scale variables
library(factoextra) # visualise kmeans clusterings
library(broom) # add clusters back to data

# For saving tables
library(kableExtra)
```

Load the initial dataset:

Convert any NULL values to na (Easier to remove in R)

```
expedia <- read_csv("data/expedia.csv", na = "NULL")
# load the initial dataset in this cell
```

Data Cleaning

Perform any data cleaning steps in this section. Insert code chunks and comments / explanations as necessary.

Extraction of variables + cleaning

1. Keep ALL useful and relevant variables for analysis

These include:

- srch_id and prop_id: Important for grouping further on
- prop_starrating: Explore impact of ratings on bookings
- gross_bookings_usd: Determine whether successful bookings occurred (na if no bookings)
- price_usd: Explore impact of pricing (Price that customers sees first)
- srch_children_count: Explore impacts of family demographics (Defined as having at least 1 child)
- comp1_inv to comp8_inv: Explores competitive categories (by availability)

2. Remove ALL na values from relevant columns

When reading csv file (expedia.csv), existing "NULL" values are converted to na. This is easier to remove in R than "NULL" value

3. Remove any bookings where no data exist for ALL competitors

Focus on bookings where competitor values are actually available

```
expedia <-
  expedia |>
  select(
    srch_id, prop_id, prop_starrating,
    gross_bookings_usd, price_usd, srch_adults_count,
```

```

srch_children_count, comp1_inv, comp2_inv,
comp3_inv, comp4_inv, comp5_inv, comp6_inv,
comp7_inv, comp8_inv # All variables required for analysis
) |>
filter(
  !is.na(gross_bookings_usd), # remove searches where bookings did not occur
  !is.na(comp1_inv & comp2_inv & comp3_inv
    & comp4_inv & comp5_inv & comp6_inv
    & comp7_inv & comp8_inv
  ) # Remove bookings where ALL competitor data is na
)

```

Tidy data (following Tidy data principles)

1. Processing some na values for competitors

Some bookings still have at least one missing competitor data. Currently, this is difficult to select certain competitors (those with na values) to remove for some bookings whilst keeping them for others.

Create a longer data frame (via `pivot_longer`) and transform `comp_inv` values to a new column "inv_value" and the respective competitor to "comp". Now filter out na values for competitors for each booking.

```

expedia_tidy <-
expedia |>
pivot_longer(
  starts_with("comp"), # selects column name starting with "comp"
  names_to = "comp",
  values_to = "inv_value"
) |>
filter(!is.na(inv_value)) # discard na values

```

Creating final analysis data frame (expedia_analysis)

1. Collapse multiple search-hotel pairs to one

By pivoting longer, we introduced multiple rows per valid booking, so we need to reduce it to a single search-hotel pair.

Grouping by both `srch_id` and `prop_id` creates rows where each row is a successful booking. The columns we need to analyse are:

- `price_usd`: To form price segmentation
- `prop_starrating`: To form hotel star rating segmentation
- `srch_children_count`: To form family segmentation
- `inv_value`: To create competitive level split

For each column, we can select the first value for each search-hotel pair.

2. Rename certain columns

To improve readability and communication of our code and dataset, the following columns are renamed:

- prop_starrating: Changed to hotel_star_ratings
- srch_children_count: Changed to children_booked
- inv_value: Changed to avg_availability

```
expedia_analysis <-  
  expedia_tidy |>  
  group_by(srch_id, prop_id) |>  
  summarise(  
    price_usd = first(price_usd),  
    hotel_star_ratings = first(prop_starrating),  
    children_booked = first(srch_children_count),  
    avg_availability = mean(inv_value, na.rm = TRUE)  
  )
```

Data Analysis

Perform your data analysis in this section. Insert code chunks and comments / explanations as necessary.

Visualise and determining split

1. Establish where the split for hotel availability should be

This can be done by observing the mean of ALL comp_inv values and the median of ALL comp_inv values. The average of the inv_values are 0.0299 and median is 0. Through this, we can conclude a natural split for:

- Either Expedia or a competitor advantage average inv_value is > 0 or < 0
- No competitive advantage average inv_value is 0

2. Check size for each split

There is 305 bookings for competitive advantage and 2836 bookings for no advantage. Reasons behind this could be the rarity of competition between the two large platforms.

The smaller dataset for competitive advantage has potential to produce less accurate results and analysis

```
# calculates mean and median for the inv_values  
expedia_competitive <-  
  expedia_tidy |>  
  summarise(  
    avg_inv = mean(inv_value),  
    median_inv = median(inv_value)  
  )  
  
# Check size for each split  
expedia_analysis |>  
  filter(avg_availability < 0 | avg_availability > 0) |> # filters either advantage for a comp  
  summarise(  
    bookings = n()  
  )  
  
expedia_analysis |>
```

```
filter(avg_availability == 0) |> # filters no competitive advantage
summarise(
  bookings = n()
)
```

Analysing first split: No advantage for Expedia or competitors

1. Extract data and prepare segmentation

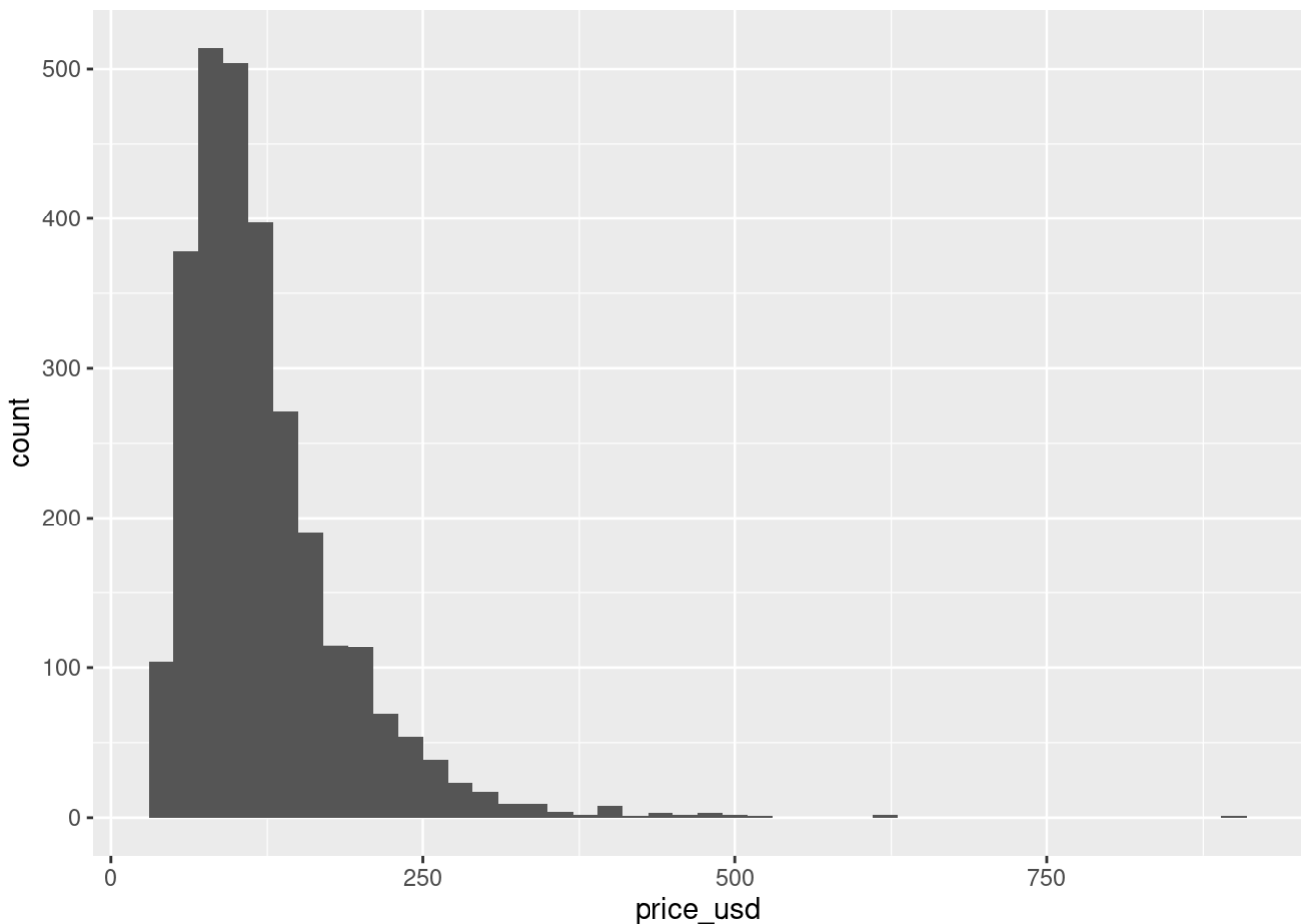
We can extract the data where no advantages exist. Then, create a visualisation to help determine where segmentations should occur for each variable

Number of "bins" used in ntiles should roughly reflect the real-world context for that variable.

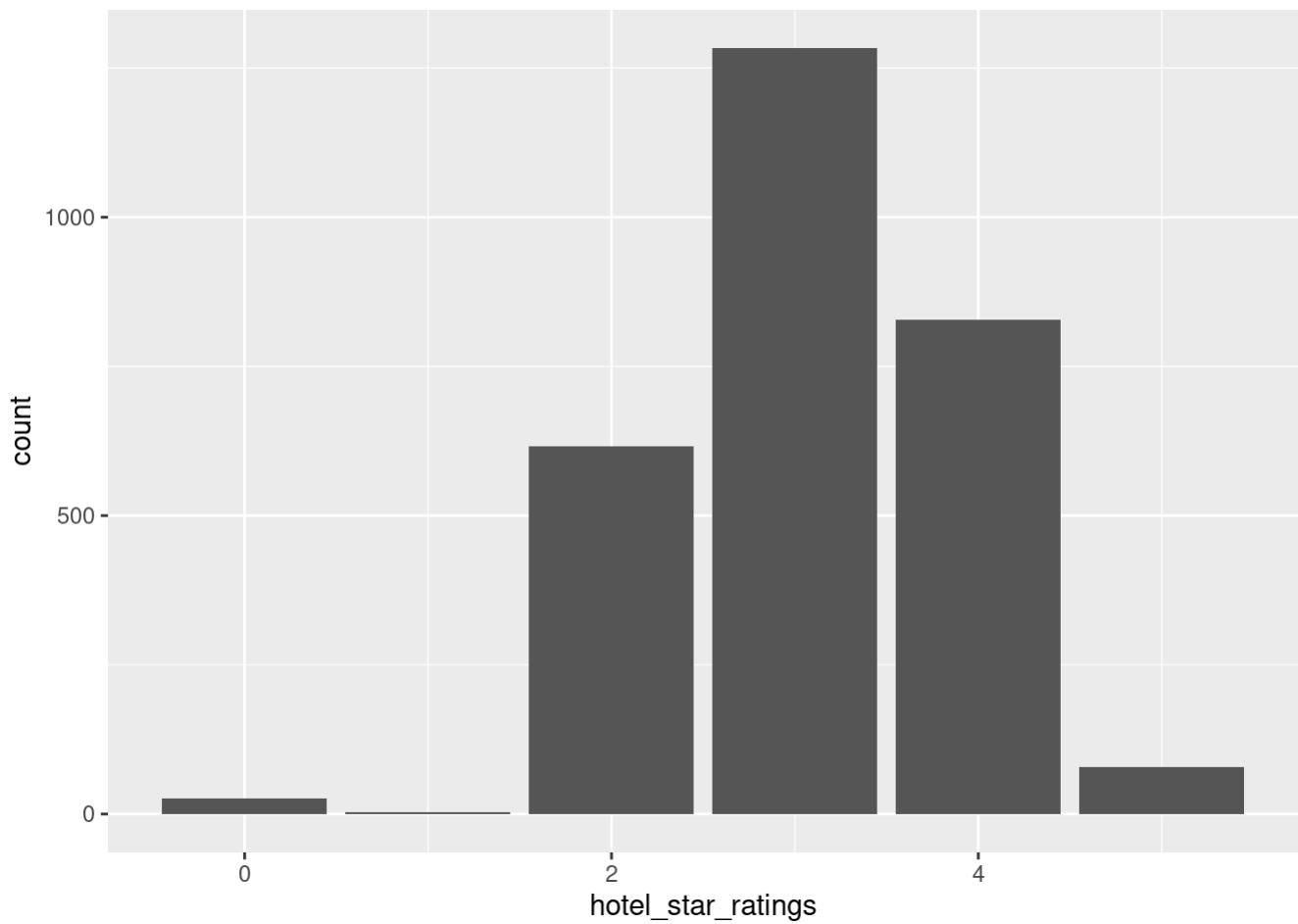
```
# extract no advantage
no_advantage_data <-
  expedia_analysis |>
  filter(avg_availability == 0)

# Visualise distribution

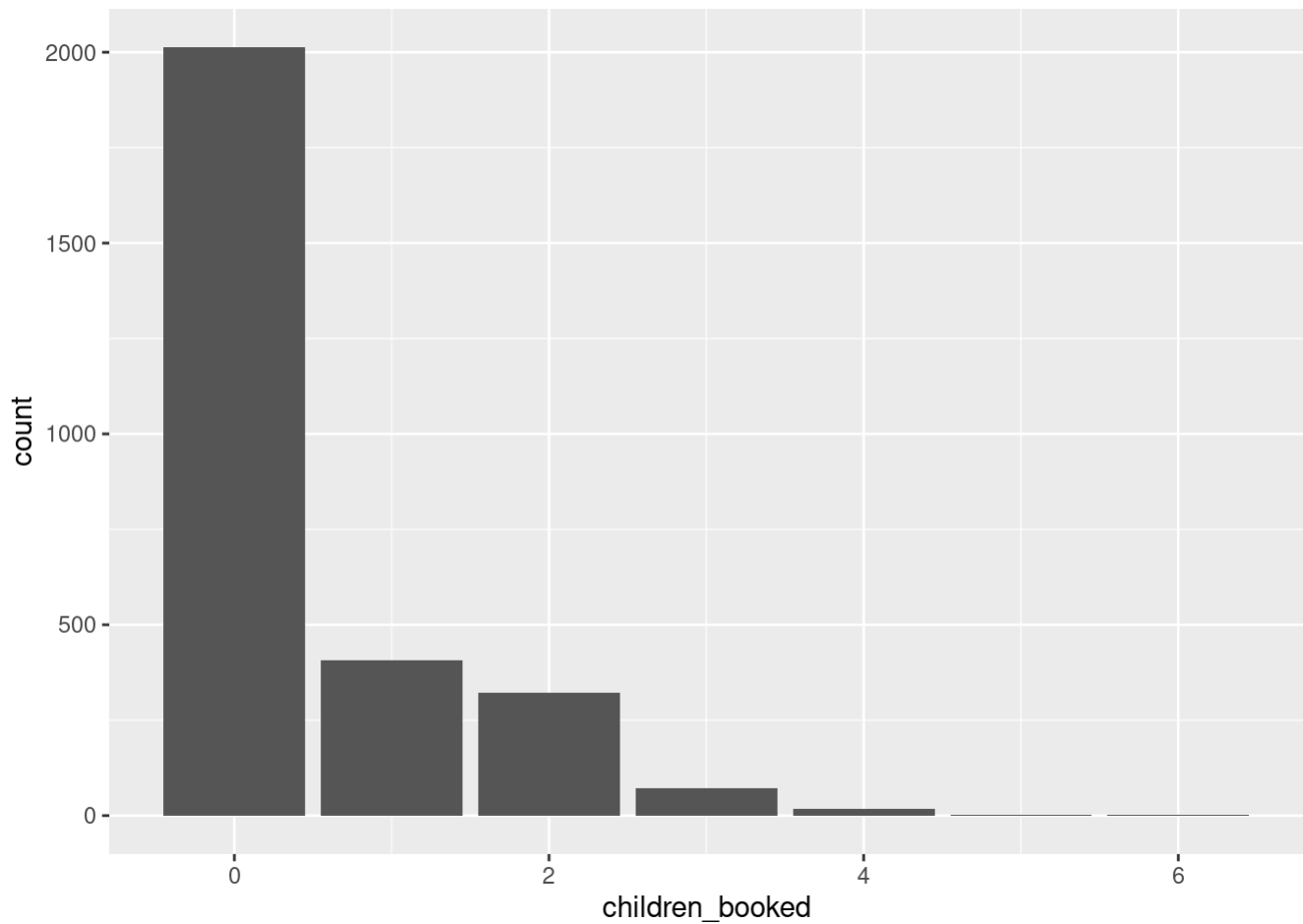
# Prices
no_advantage_data |>
  ggplot(aes(x = price_usd)) +
  geom_histogram(binwidth = 20) # Range of 20 for each bin to effectively cover price range
```



```
# Hotel star ratings
no_advantage_data |>
  ggplot(aes(x = hotel_star_ratings)) +
  geom_bar() # bar graph for categorical variable
```



```
# Families
no_advantage_data |>
  ggplot(aes(x = children_booked)) +
  geom_bar() # bar graph for categorical variable
```



```
# Prep for segmenting

# Hotel star ratings:

# Only 29 bookings are at 0 or 1 star ratings
no_advantage_data |>
  filter(hotel_star_ratings <= 1) |>
  count()
```

```
# A tibble: 29 × 2
# Groups:   srch_id [29]
  srch_id      n
  <dbl> <int>
1    2716     1
2    3790     1
3    8146     1
4   14092     1
5   15343     1
6   22945     1
7   26643     1
8   28194     1
9   29414     1
10  30819     1
# i 19 more rows
```

```
# Remove 0 and 1 star rating to prevent unwanted effects on final analysis
no_advantage_data <-
  no_advantage_data |>
  filter(hotel_star_ratings >= 2)

# Family:
no_advantage_data |>
  filter(children_booked == 0) |>
  count()
```

```
# A tibble: 1,996 × 2
# Groups:   srch_id [1,996]
  srch_id      n
  <dbl> <int>
1     297     1
2     325     1
3     431     1
4     547     1
5     606     1
6     669     1
7     780     1
8     935     1
9    1032     1
10   1155     1
# i 1,986 more rows
```

```
# 811 no family vs 1996 with family
```

Prices - The price values are heavily right skewed, so there is not a normal distribution. By using ntiles, we can reliably create accurate segmentation for each price range (Low, medium and high).

Hotel star rating - There is a minimal amount of 0 and 1 star rating hotels booked (29 bookings), the majority are 3 star rated. We can use hard cutoffs (via case_when to decide on the segmentation with 2 stars, 3 stars and 4 or more stars).

0 and 1 star bookings are removed to prevent unwanted impact on final analysis since there is such a small amount of bookings, it acts as outliers.

Family - The large majority of bookings are by non-families. Using ntiles is unsuitable since it would redistribute all values. So case_when() can be used again to create cutoffs which exist naturally (0 for non-family and >= 1 for families).

2. Construction of segmentation

```
no_advantage_final <-
  no_advantage_data |>
  ungroup() |>
  mutate(
    ptile = ntile(price_usd, 3), # create 3 tiles for price
    price_segment = case_when(
      ptile == 1 ~ "Low Price",
      ptile == 2 ~ "Middle Price",
```

```

    ptile == 3 ~ "High Price"
  ),
  rating_segment = case_when ( # manually creating segments
    hotel_star_ratings == 2 ~ "Low Rating",
    hotel_star_ratings == 3 ~ "Middle Rating",
    hotel_star_ratings >= 4 ~ "High Rating"
  ),
  family = case_when( # manually creating segments
    children_booked == 0 ~ "No children",
    children_booked >= 1 ~ "Family"
  )
)

```

Verify successful segments

```

no_advantage_final |>
  count(price_segment) # should have roughly equal groups

no_advantage_final |>
  count(rating_segment) # should reflect actual distribution of ratings (unequal for segments)

no_advantage_final |> # should reflect actual distribution of family demographics (unequal for
  count(family)

```

3. Create summary table

```

no_advantage_summary <-
no_advantage_final |>

# Group different combinations of segments
group_by(price_segment, rating_segment, family) |>
summarise(
  bookings = n(), # count number of bookings
  avg_price = mean(price_usd), # find average price of hotel
  avg_rating = mean(hotel_star_ratings), # find average star ratings
  avg_children = mean(children_booked), # find average number of children
  .groups = "drop"
)

# Create neater table to output
no_advantage_summary |>
  kable(digits = 2)

```

price_segment	rating_segment	family	bookings	avg_price	avg_rating	avg_children
High Price	High Rating	Family	172	202.58	4.10	1.55
High Price	High Rating	No children	398	214.41	4.15	0.00
High Price	Low Rating	Family	13	167.46	2.00	1.85
High Price	Low Rating	No children	25	167.76	2.00	0.00
High Price	Middle Rating	Family	125	166.94	3.00	1.62

price_segment	rating_segment	family	bookings	avg_price	avg_rating	avg_children
High Price	Middle Rating	No children	202	168.67	3.00	0.00
Low Price	High Rating	Family	18	78.28	4.00	1.89
Low Price	High Rating	No children	82	73.07	4.01	0.00
Low Price	Low Rating	Family	98	68.08	2.00	1.85
Low Price	Low Rating	No children	332	65.45	2.00	0.00
Low Price	Middle Rating	Family	110	75.28	3.00	1.56
Low Price	Middle Rating	No children	296	72.78	3.00	0.00
Middle Price	High Rating	Family	58	110.16	4.00	1.64
Middle Price	High Rating	No children	179	112.46	4.01	0.00
Middle Price	Low Rating	Family	42	104.88	2.00	1.64
Middle Price	Low Rating	No children	106	104.49	2.00	0.00
Middle Price	Middle Rating	Family	175	108.85	3.00	1.69
Middle Price	Middle Rating	No children	376	110.06	3.00	0.00

4. Comparisons and analysis

Construct a rough plot to visualise general patterns. Then observe 3 specific arrangements of individual tables (to analyse more specific patterns). These tables are:

- Relationship between Pricing and family
- Relationship between Ratings and family
- Relationship between Pricing and Ratings

```
# Have a rough visualisation

no_advantage_plot <-
no_advantage_summary |>
  ggplot(aes(x = price_segment, y = bookings, fill = rating_segment)) +

  # Places overlapping objects directly besides each other; easier to visualise patterns
  geom_col(position = "dodge") +
  facet_wrap(~ family) +

  # Colour blindness
  scale_fill_okabe_ito() +

  # Rename title, x-axis title, y-axis title and add subtitle
  labs(
    title = "Number of bookings by pricing, rating and family factor",
    subtitle = "For no competitive advantage",
    x = "Price Segments",
    y = "Number of bookings",
    fill = "Rating Segments" # Legend for fill
  ) +
  theme_bw() +

  # Format x-axis labels for easier viewing
```

```
theme(axis.text.x = element_text(angle = 45, hjust = 1, vjust = 1))

# Save plot to render later
ggsave("no_advantage_plot.png", plot = no_advantage_plot)

no_advantage_plot
```



General Analysis:

- 39.9% of all bookings are from highly priced and rated hotels and medium priced and rated hotels. Suggests price and rating of hotels have a strong linear relationship
- Non-family bookings are typically highly priced and rated hotels rather than medium priced and rated
- Family bookings are more even between highly priced and rated hotels and medium priced and rated
- Low rated hotels are consistently booked less, regardless of price

a. Relationship between Pricing and Rating

```
no_advantage_summary |>
  group_by(price_segment, rating_segment) |> # create segment pairing
  summarise(
    bookings = sum(bookings), # find total bookings
    avg_price = mean(avg_price), # find average price spent
    .groups = "drop"
  ) |>
```

```
arrange(desc(bookings)) |>
kable(digits = 2)
```

price_segment	rating_segment	bookings	avg_price
High Price	High Rating	570	208.50
Middle Price	Middle Rating	551	109.45
Low Price	Low Rating	430	66.76
Low Price	Middle Rating	406	74.03
High Price	Middle Rating	327	167.80
Middle Price	High Rating	237	111.31
Middle Price	Low Rating	148	104.69
Low Price	High Rating	100	75.68
High Price	Low Rating	38	167.61

General Analysis:

- There exists a strong relationship between the level of rating, pricing, and bookings such that highly priced and rated hotels are booked the most
- Price might be considered more by customers. Out of all price segments, low priced hotels are both in the top 5 most booked (Needs further analysis to confirm relation)

b. Relationship between Price and Family

```
no_advantage_summary |>
  group_by(price_segment, family) |> # create segment pairing
  summarise(
    bookings = sum(bookings), # find total bookings per segment pair
    avg_price = mean(avg_price), # find average price
    .groups = "drop"
  ) |>
  arrange(desc(bookings)) |>
  kable(digits = 2)
```

price_segment	family	bookings	avg_price
Low Price	No children	710	70.43
Middle Price	No children	661	109.00
High Price	No children	625	183.61
High Price	Family	310	178.99
Middle Price	Family	275	107.96
Low Price	Family	226	73.88

General Analysis:

- 35.6% of bookings by non-families are at low priced hotels, 27.9% of bookings by families at low priced hotels
- Could suggest non-families are more hesitant on spending

- This is evident by low priced bookings being the most for non-families compared to high priced bookings being the most for families

c. Relationship between Rating and Family

```
no_advantage_summary |>
  group_by(rating_segment, family) |> # create segment pairing
  summarise(
    bookings = sum(bookings), # find total booking per segment pair
    avg_price = mean(avg_price), # find average price
    .groups = "drop"
  ) |>
  arrange(desc(bookings)) |>
  kable(digits = 2)
```

rating_segment	family	bookings	avg_price
Middle Rating	No children	874	117.17
High Rating	No children	659	133.32
Low Rating	No children	463	112.57
Middle Rating	Family	410	117.02
High Rating	Family	248	130.34
Low Rating	Family	153	113.47

General Analysis:

- Both family and non-family demographics have same ranking for rating segment
- Middle rated hotels are most booked for both, followed by high rating and low rating
- Suggests family demographics has a very minimal correlation to ratings of a booking but does require more detail analysis to confirm.

5. Kmeans segmentation (For further analysis)

First, re-scale ALL required variables through standardisation. This creates normal distributions which is needed for accurate cluster constructions.

Otherwise, ONLY the variable with most variation will determine cluster formation

```
no_advantage_recipe <-
  recipe(~ price_usd + hotel_star_ratings + children_booked,
    data = no_advantage_data
  ) |>
  step_normalize(all_numeric_predictors()) # create normally distributed variables

no_advantage_scaled <-
  no_advantage_recipe |>
  prep() |>
  bake(new_data = NULL)
```

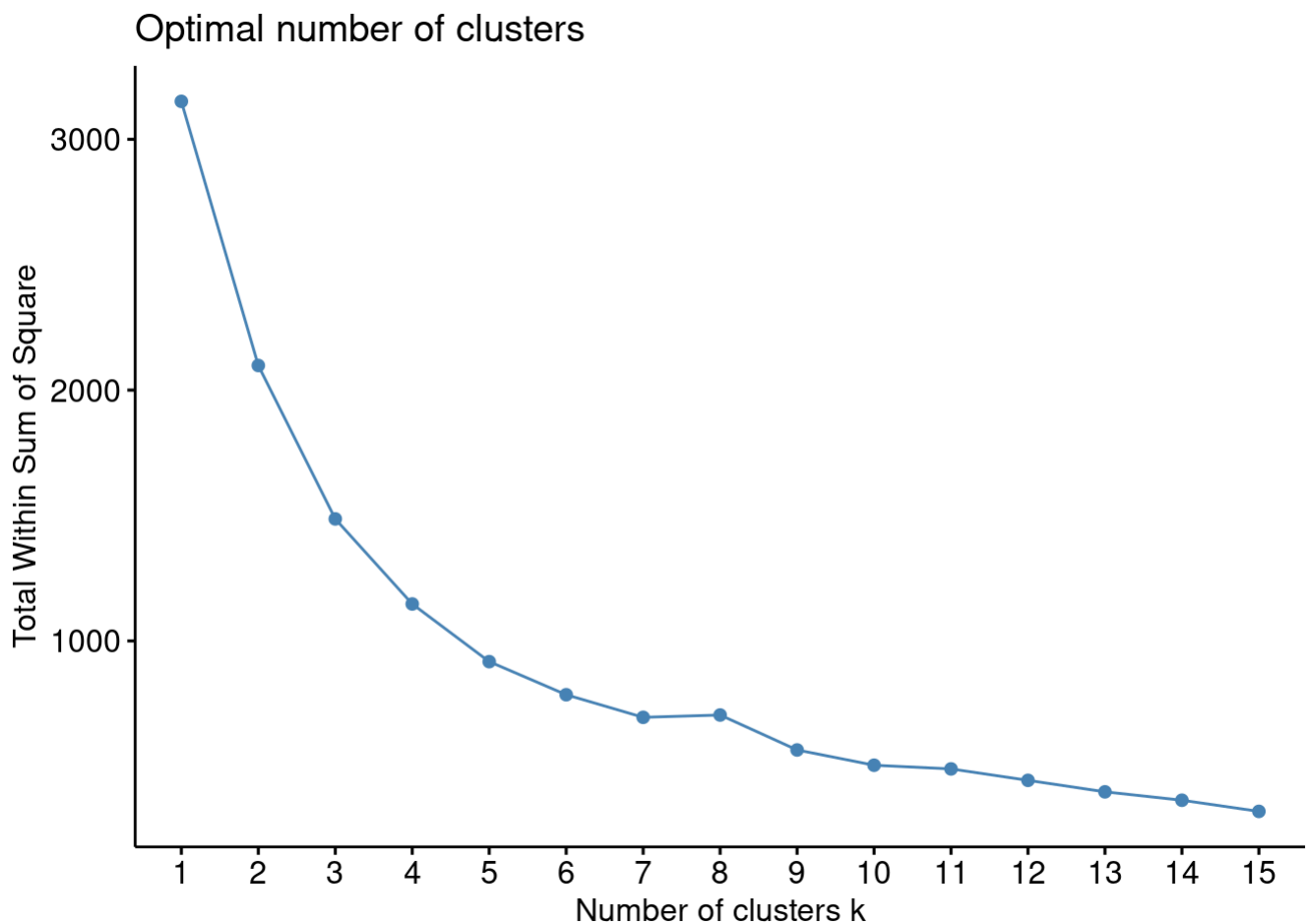
6. Determine ideal number of clusters

Use a sample of our available data to allow R to calculate a number of different clusters for us to decide on.

The elbow method allows us to choose an "optimal" number of clusters by observing where the plot begins to flatten. This indicates all clusters afterwards would not vary too much and so we can select an optimal number of clusters.

```
set.seed(42) # initialise random number generator to maintain consistent results
no_advantage_sample <-
  no_advantage_scaled |>
  sample_n(1000) # select 1000 bookings

fviz_nbclust(
  no_advantage_sample,
  kmeans,
  method = "wss",
  k.max = 15, # maximum clusters of 15
  nstart = 5 # use 5 different starts
)
```



Although the elbow appears to be at 7, repeated changing and observing cluster plot shows excessive overlapping clusters unless cluster number is set to 3. Thus, through trial and error, the optimal cluster selected is 3.

This resulted in almost no overlapping clusters, so we can conclude this number of cluster will present the most accurate results.

7. Construct kmeans clustering

Our “optimal” number of clusters is 3. By assigning this to centers, R will choose 3 random points as the “middle”. It then measures the distance between each point to those numbers and shuffles to create new centres as well as reassigning each point.

Finally, after a maximum number of shuffles or the centres does not change, this becomes how each clusters can be formed.

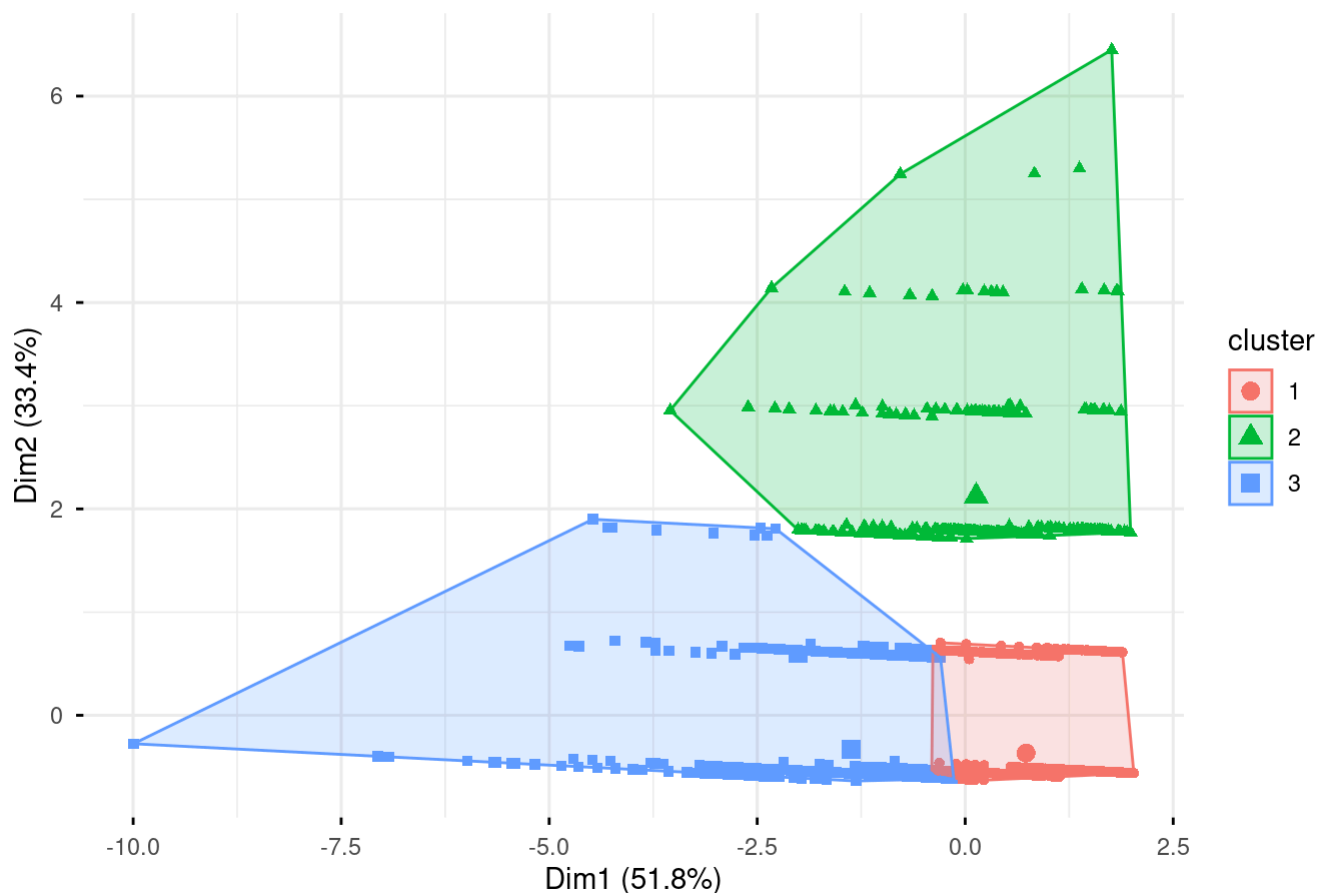
```
set.seed(420) # initialise random number generator to ensure consistent results
kmeans_3 <-
  kmeans(no_advantage_scaled,
         centers = 3) # use optimal clusters of 3
```

```
# Check kmeans clusters
kmeans_3
```

Visualise kmeans clusters

```
fviz_cluster(
  kmeans_3,
  data = no_advantage_scaled,
  geom = "point",
  ggtheme = theme_minimal()
)
```

Cluster plot



All 3 clusters are very distinct and offers minimal to no overlaps. This is positive for our analysis as we would like to observe more specific and hidden patterns that has not been revealed by rule-of-thumbs segmentations.

8. Add kmeans clusters to original data with rule of thumb segments

```
no_advantage_clusters <-
  augment(kmeans_3, no_advantage_final)

head(no_advantage_clusters)
```

A tibble: 6 × 11

```
srch_id prop_id price_usd hotel_star_ratings children_booked avg_availability
  <dbl>   <dbl>   <dbl>         <dbl>         <dbl>         <dbl>
1    118   86727     99             3             2             0
2    297   38088    167             2             0             0
3    325   41261    134             3             0             0
4    431   98457    259             4             0             0
5    523   45995     99             2             1             0
6    547  140801     89             3             0             0
```

```
# i 5 more variables: ptile <int>, price_segment <chr>, rating_segment <chr>,
#   family <chr>, .cluster <fct>
```

9. Create summary table

```
no_advantage_clust_summary <-
  no_advantage_clusters |>
  group_by(.cluster) |>
  summarise(
    bookings = n(),
    avg_price = mean(price_usd),
    avg_rating = mean(hotel_star_ratings),
    avg_children_booked = mean(children_booked),
    .groups = "drop"
  ) |>
  rename(Clusters = .cluster) |>
  arrange(desc(bookings))

no_advantage_clust_summary <-
  no_advantage_clust_summary |>
  kable(digits = 2)

no_advantage_clust_summary
```

Clusters	bookings	avg_price	avg_rating	avg_children_booked
1	1544	95.26	2.67	0.16
3	863	179.26	4.00	0.20
2	400	119.42	3.05	2.30

General Analysis:

- 55.0% of bookings come from cluster 1. This indicates the majority of customers book cheaper and lower rated hotels. Suggests pricing is more significant for customers
- 14.3% of bookings come from cluster 2. The higher average children booked suggest this is our family cluster. Suggests targeting non-family bookings are more efficient than focusing on the family demographic

Analysing second split: Availability advantage for Expedia OR competitors

1. Data extraction and segmentation preparation

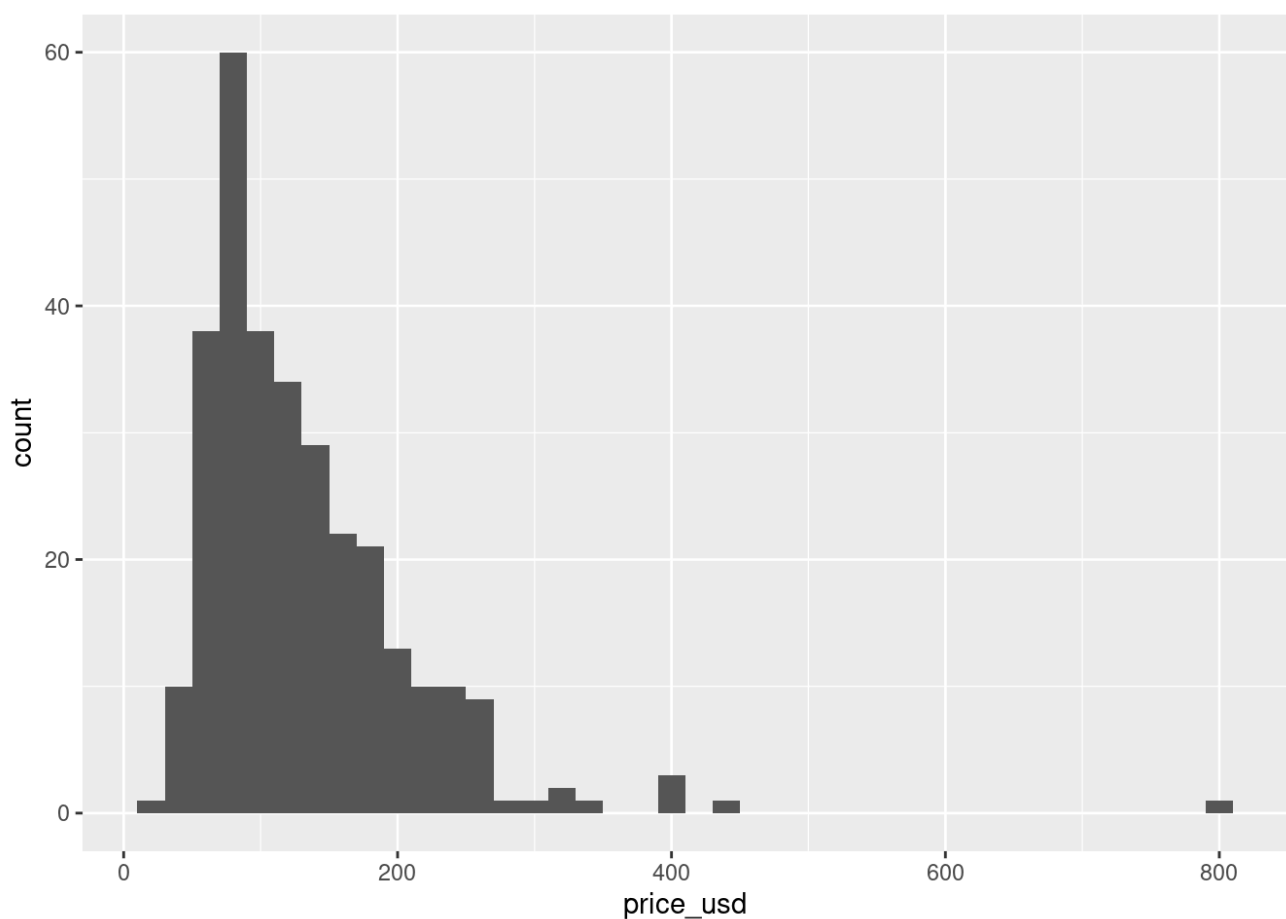
Extract data for where an availability advantage exists (this is for Expedia or another competitor). Indicated by `avg_availability > 0` (Expedia advantage) or `< 0` (Competitor advantage)

Create visualisation to understand where segmentation should be formed to approximate natural segments of data set and variables.

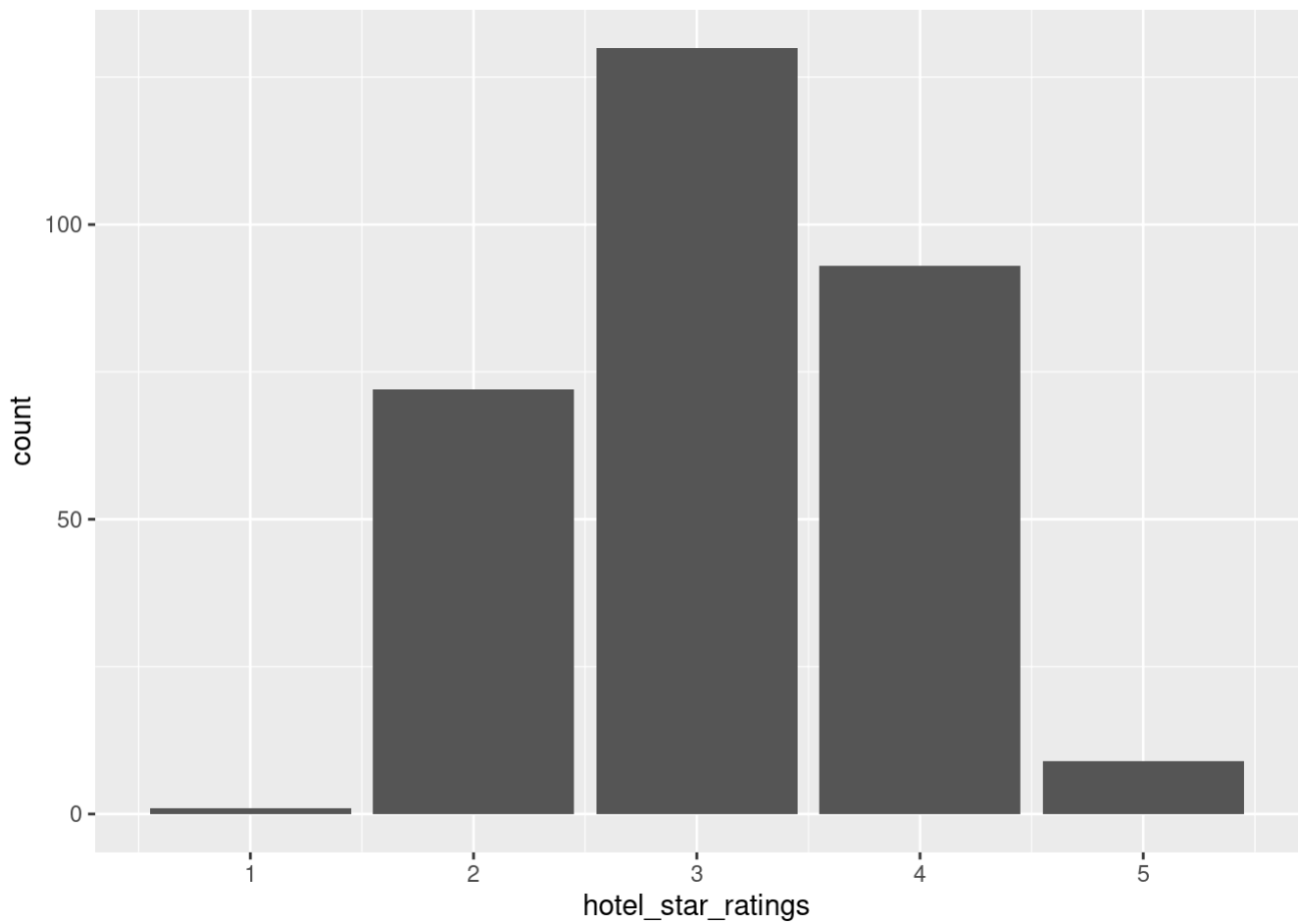
```
# Extract availability advantage data
advantage_data <-
  expedia_analysis |>
  filter(avg_availability > 0 | avg_availability < 0)

# Visualise distribution

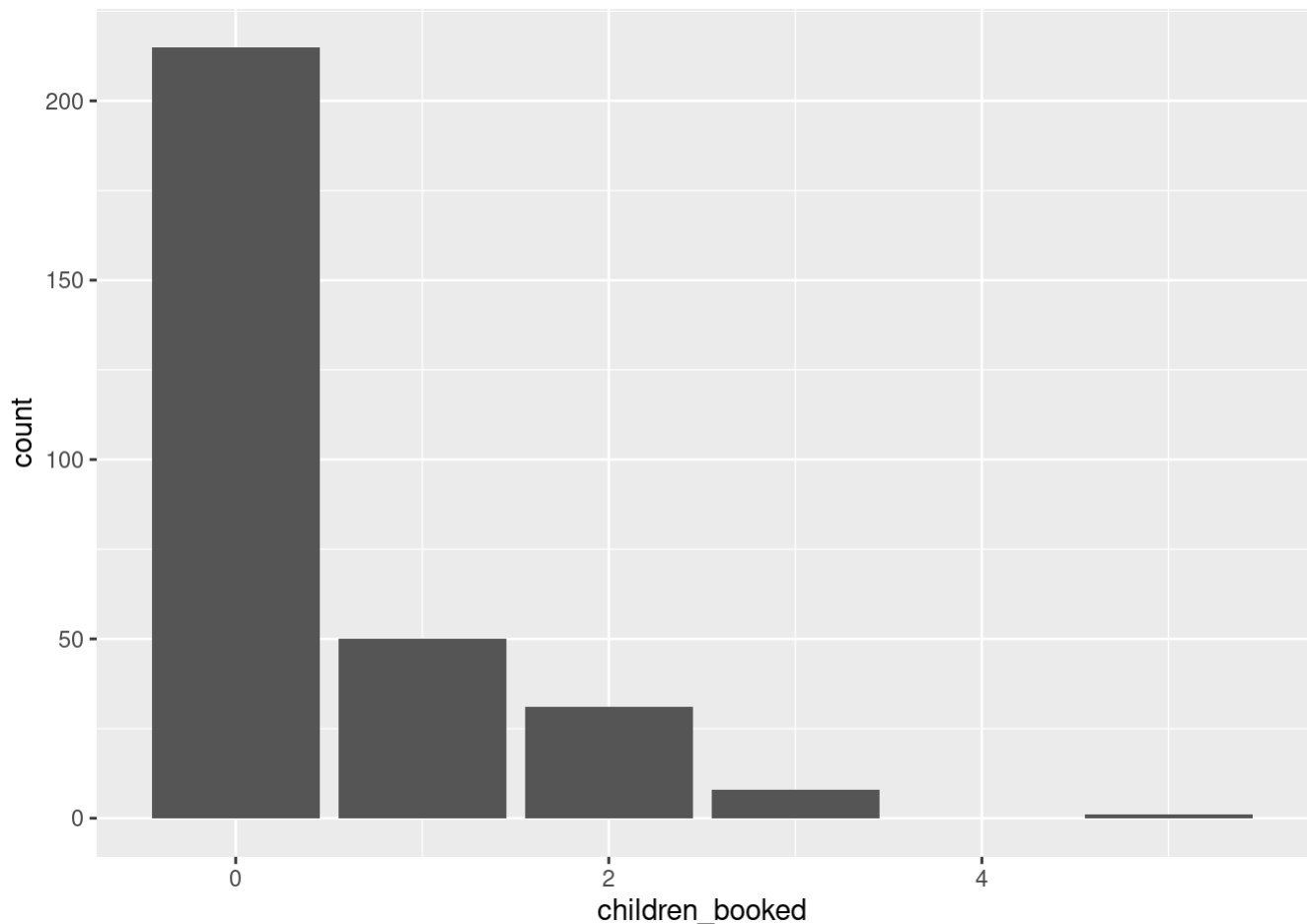
# Prices
advantage_data |>
  ggplot(aes(x = price_usd)) +
  geom_histogram(binwidth = 20) # range of 20 for each bins
```



```
# Hotel star ratings  
advantage_data |>  
  ggplot(aes(x = hotel_star_ratings)) +  
  geom_bar()
```



```
# Family  
advantage_data |>  
  ggplot(aes(x = children_booked)) +  
  geom_bar()
```



```
# Prep for segmenting
```

```
# Hotel star rating:
```

```
advantage_data |>
```

```
  filter(hotel_star_ratings == 1) |> # Find number of 1 star rating bookings  
  count()
```

```
# A tibble: 1 × 2
```

```
# Groups:   srch_id [1]
```

```
  srch_id      n  
    <dbl> <int>
```

```
1  91867      1
```

```
advantage_data <-
```

```
  advantage_data |>
```

```
  filter(hotel_star_ratings >= 2) # keep bookings with at least 2 stars
```

```
# Family:
```

```
# Count number of non-families vs families
```

```
advantage_data |>
```

```
  filter(children_booked == 0) |>  
  count()
```

```
# A tibble: 215 × 2
```

```
# Groups:   srch_id [215]
```

```
  srch_id      n
```

```

      <dbl> <int>
1      1027      1
2      1447      1
3      2594      1
4      2615      1
5      2654      1
6      2740      1
7      2900      1
8      3174      1
9      3258      1
10     3888      1
# i 205 more rows

```

```

advantage_data |>
  filter(children_booked >= 1) |>
  count()

```

```

# A tibble: 89 × 2
# Groups:   srch_id [89]
  srch_id      n
  <dbl> <int>
1     5683      1
2     6157      1
3     7707      1
4     8057      1
5     8918      1
6     9063      1
7    10305      1
8    11831      1
9    13013      1
10   13920      1
# i 79 more rows

```

```
# 215 non-families booking vs 89 family bookings
```

Prices - There is a high concentration of bookings that occurs at the lower price range and very minimal outliers that exist above. Using ntiles, the prices can be dropped into each bin so that each segment is roughly equal.

Hotel star ratings - The distribution of hotel star ratings are more compact than the previous selection of data (perhaps due to a smaller dataset). We can use case_when() function to manually determine the segments for each rating.

Also, remove bookings below 2 stars since there are only 1, making it an outlier and would likely introduce unwanted impact on analysis

Family - The majority of bookings are of non-families, again could potentially be of the same reason as the first dataset. 215 are non-family bookings and 89 are family bookings.

Use case_when() and set family as children_booked >= 1 and non-family as children_booked == 0.

2. Segmentation construction

```
# Code copied from no_advantage_final analysis
advantage_final <-
  advantage_data |>
  ungroup() |>
  mutate(
    ptile = ntile(price_usd, 3),
    price_segment = case_when(
      ptile == 1 ~ "Low Price",
      ptile == 2 ~ "Middle Price",
      ptile == 3 ~ "High Price"
    ),
    rating_segment = case_when (
      hotel_star_ratings == 2 ~ "Low Rating",
      hotel_star_ratings == 3 ~ "Middle Rating",
      hotel_star_ratings >= 4 ~ "High Rating"
    ),
    family = case_when(
      children_booked == 0 ~ "No children",
      children_booked >= 1 ~ "Family"
    )
  )
```

```
# Verify successful segments
advantage_final |>
  count(price_segment)

advantage_final |>
  count(rating_segment)

advantage_final |>
  count(family)
```

3. Create summary table

```
# Code copied from no_advantage_final analysis
advantage_summary <-
  advantage_final |>
  group_by(price_segment, rating_segment, family) |>
  summarise(
    bookings = n(), # sum number of bookings
    avg_price = mean(price_usd), # find average price
    avg_rating = mean(hotel_star_ratings), # find average star rating
    avg_children = mean(children_booked), # find average number of children booked
    .groups = "drop"
  )

# Nicer looking table via kable()
advantage_summary |>
  kable(digits = 2)
```

price_segment	rating_segment	family	bookings	avg_price	avg_rating	avg_children
High Price	High Rating	Family	16	225.38	4.06	1.19
High Price	High Rating	No children	45	235.40	4.18	0.00
High Price	Low Rating	Family	5	171.80	2.00	1.60
High Price	Low Rating	No children	4	200.75	2.00	0.00
High Price	Middle Rating	Family	7	199.43	3.00	2.14
High Price	Middle Rating	No children	24	192.71	3.00	0.00
Low Price	High Rating	Family	5	62.60	4.00	1.60
Low Price	High Rating	No children	13	75.77	4.00	0.00
Low Price	Low Rating	Family	18	67.17	2.00	1.56
Low Price	Low Rating	No children	32	66.78	2.00	0.00
Low Price	Middle Rating	Family	11	75.27	3.00	1.64
Low Price	Middle Rating	No children	23	73.04	3.00	0.00
Middle Price	High Rating	Family	4	129.50	4.00	1.00
Middle Price	High Rating	No children	19	116.26	4.00	0.00
Middle Price	Low Rating	Family	4	123.75	2.00	2.50
Middle Price	Low Rating	No children	9	114.00	2.00	0.00
Middle Price	Middle Rating	Family	19	113.16	3.00	1.58
Middle Price	Middle Rating	No children	46	117.63	3.00	0.00

4. Comparisons and Analysis

Create rough visualisation to evaluate any particular patterns or general analysis. Then, on a much deeper scale, we can observe the individual patterns on a two-variable level. This forms tables of:

- Relationship between Pricing and family
- Relationship between Ratings and family
- Relationship between Pricing and Ratings

```

advantage_plot <-
advantage_summary |>
  ggplot(aes(x = price_segment, y = bookings, fill = rating_segment)) +

  # Places overlapping objects directly besides each other; easier to visualise patterns
  geom_col(position = "dodge") +
  facet_wrap(~ family) +

  # Colour blindness
  scale_fill_okabe_ito() +
  # create title, x-axis title, y-axis title and a subtitle
  labs(
    title = "Number of bookings by pricing, rating and family factor",
    subtitle = "For any competitive advantage",
    x = "Price Segments",
    y = "Number of bookings",
  )

```

```

fill = "Rating Segments" # Change name for Ratings key
) +
theme_bw() +

# Angle x-axis labels for easier viewing
theme(axis.text.x = element_text(angle = 45, hjust = 1, vjust = 1))

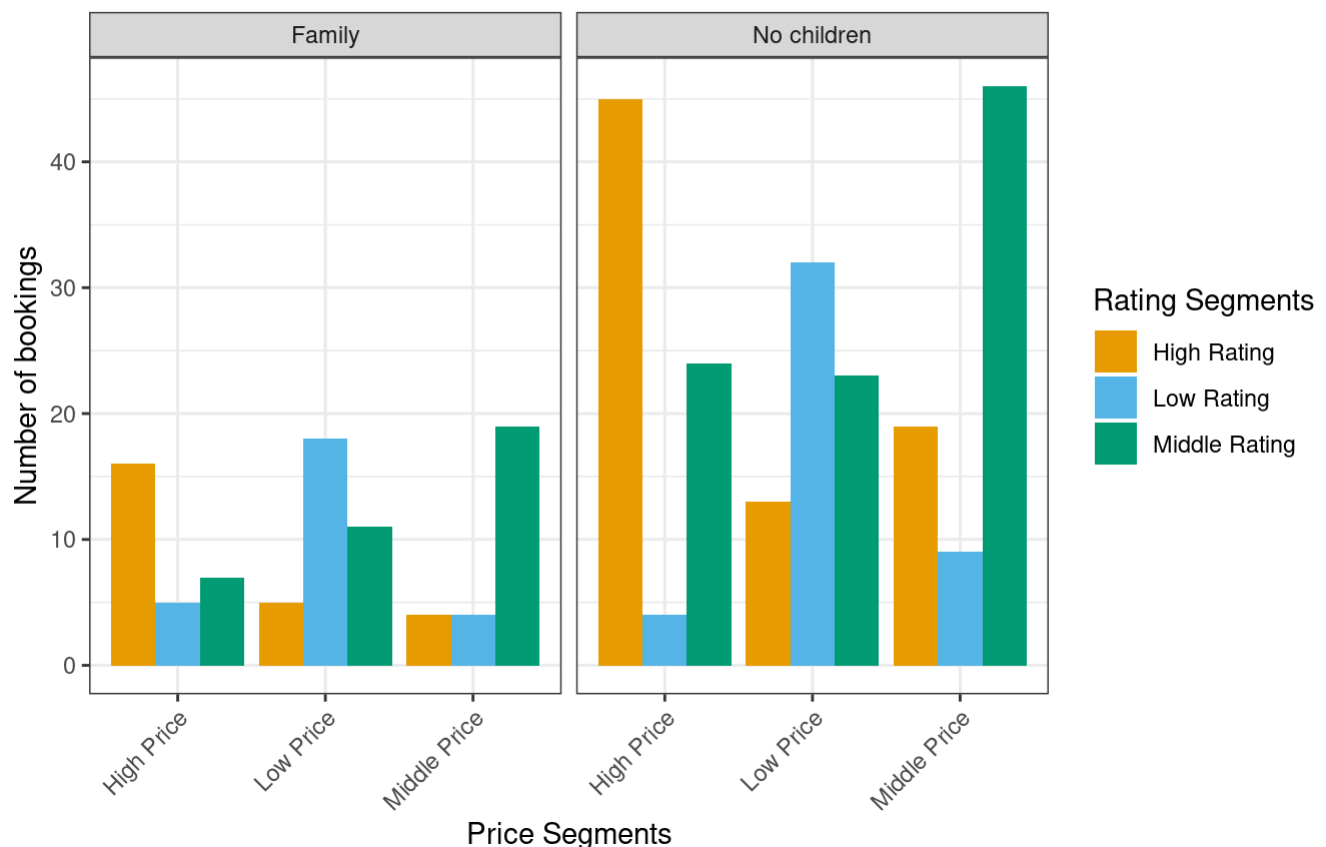
ggsave("advantage_plot.png", plot = advantage_plot)

advantage_plot

```

Number of bookings by pricing, rating and family factor

For any competitive advantage



General Analysis:

- Across both family and non-family demographics, middle priced and rated bookings are the highest
- This indicates price could still be the most significant factor as even though availability varies, customers does not "splurge"
- Low priced and rated hotels forms a higher proportion in competitor advantage context. Suggests less availability results in more lower quality hotel bookings

a. Relationship between Pricing and Family

```

# Code copied from no_advantage analysis
advantage_summary |>

# create price family pairs
group_by(price_segment, family) |>

```

```

summarise(
  bookings = sum(bookings), # calculate number of bookings
  avg_price = mean(avg_price), # find average price
  .groups = "drop"
) |>
arrange(desc(bookings)) |> # arrange from most bookings first
kable(digits = 2)

```

price_segment	family	bookings	avg_price
Middle Price	No children	74	115.96
High Price	No children	73	209.62
Low Price	No children	68	71.86
Low Price	Family	34	68.35
High Price	Family	28	198.87
Middle Price	Family	27	122.14

General Analysis:

- Middle priced bookings make up 33.2% of all bookings when competitive advantage exists
- Suggests pricing is still the most significant indicator of bookings
- However, it could also result from middle priced rooms having the most availability (compared to high priced which would be booked out first due to a variety of reasons)

b. Relationship between Pricing and Rating

```

# Code copied from no_advantage analysis
advantage_summary |>

# create price and rating pairs
group_by(price_segment, rating_segment) |>
summarise(
  bookings = sum(bookings), # calculate total bookings
  avg_price = mean(avg_price), # find average price
  .groups = "drop"
) |>
arrange(desc(bookings)) |> # arrange with most bookings first
kable(digits = 2)

```

price_segment	rating_segment	bookings	avg_price
Middle Price	Middle Rating	65	115.39
High Price	High Rating	61	230.39
Low Price	Low Rating	50	66.97
Low Price	Middle Rating	34	74.16
High Price	Middle Rating	31	196.07
Middle Price	High Rating	23	122.88
Low Price	High Rating	18	69.18

price_segment	rating_segment	bookings	avg_price
Middle Price	Low Rating	13	118.88
High Price	Low Rating	9	186.28

General Analysis:

- Once again, middle priced and rated bookings and high priced and rated bookings have the most bookings
- Interestingly, the remainder are ranked similar to no competitive advantage. Suggests booking patterns are decently stable
- 20.3% of bookings are high priced and rated bookings for no competitive advantage and 20.1% of bookings are high priced and rated bookings for competitive advantage
- Further reinforces stability of customer preferences for bookings

c. Relationship between Rating and Family

```

advantage_summary |>

# create rating and family pairs
group_by(rating_segment, family) |>
summarise(
  bookings = sum(bookings), # calculate total number of bookings
  avg_price = mean(avg_price), # find average price
  .groups = "drop"
) |>
arrange(desc(bookings)) |> # arrange with most bookings first
kable(digits = 2)

```

rating_segment	family	bookings	avg_price
Middle Rating	No children	93	127.79
High Rating	No children	77	142.48
Low Rating	No children	45	127.18
Middle Rating	Family	37	129.29
Low Rating	Family	27	120.91
High Rating	Family	25	139.16

General Analysis:

- Only difference from competitive advantage context is high rating family bookings being the lowest rather than low rating family bookings for no competitive advantage
- Can further prove our hypothesis that availability advantage has no significant impact on bookings or behaviours

5. Kmeans segmentations (For deeper analysis)

Standardise the 3 variables required to form normal distributions. This ensures that the price, hotel rating and family variables are considered when determining cluster splits.

If this is not done, R will take the variable with the most variations and base the split of that singular variable.

```
advantage_recipe <-  
  recipe(~ price_usd + hotel_star_ratings + children_booked,  
          data = advantage_data) |>  
  step_normalize(all_numeric_predictors())  
  
advantage_scaled <-  
  advantage_recipe |>  
  prep() |>  
  bake(new_data = NULL)
```

6. Determine ideal number of clusters

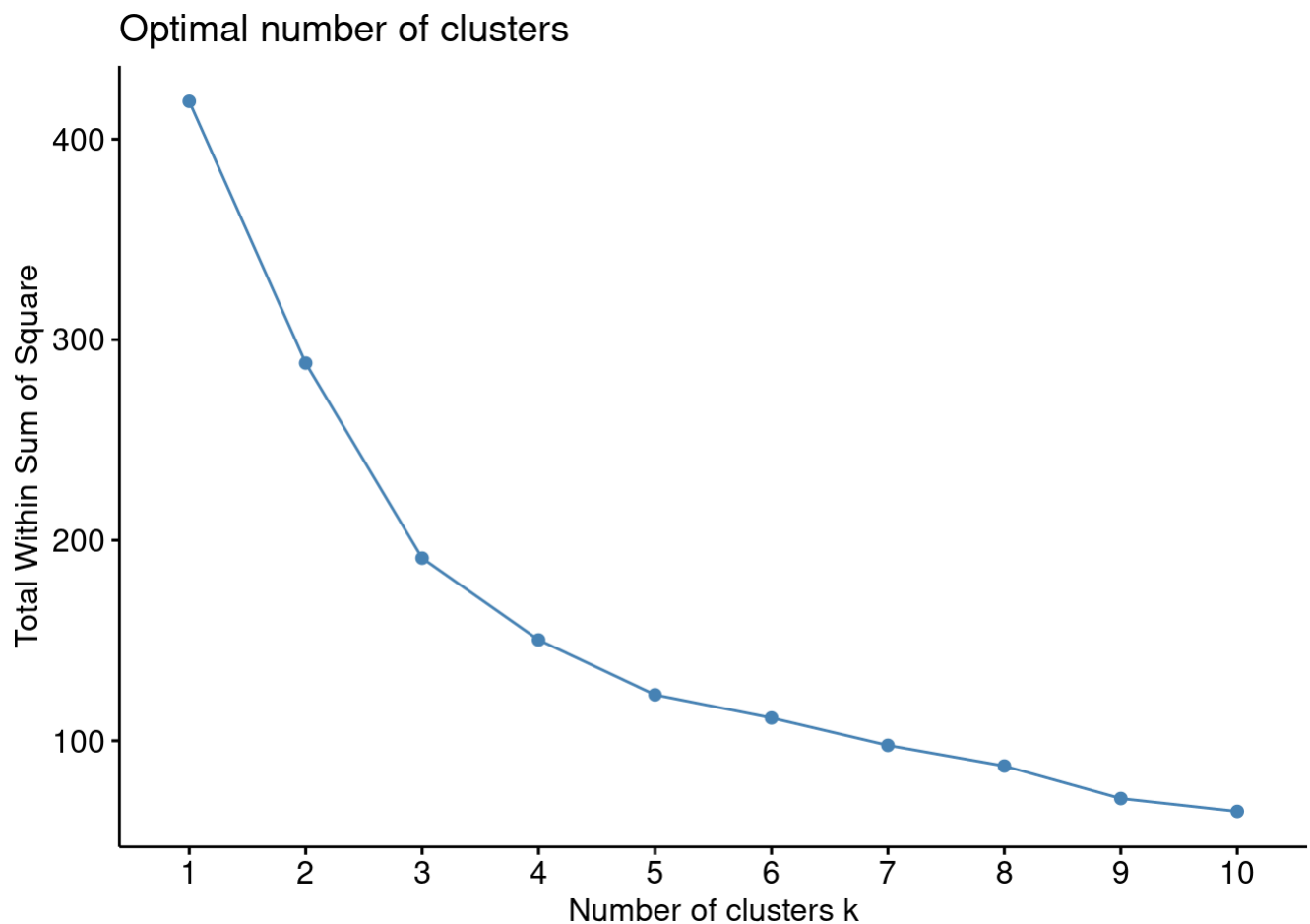
Choose a sample from the dataset as data for R to figure out a range of clusters to select the most optimal.

The elbow method allows us to choose an “optimal” number of clusters by observing where the plot begins to flatten. This indicates all clusters afterwards would not vary too much and so we can select an optimal number of clusters.

```
set.seed(42) # initialise random number generator to produce consistent results  
advantage_sample <-  
  advantage_scaled |>  
  sample_n(150) # due to smaller dataset, sample size is reduced
```

7. Visualise and determine optimal clusters (using elbow method)

```
fviz_nbclust(  
  advantage_sample,  
  kmeans,  
  method = "wss",  
  k.max = 10,  
  nstart = 5  
)
```



Although initially the optimal number of clusters appears to be 5 (as values flatten afterwards), through trial and error and observing the plot for least significant overlaps, the most optimal number of clusters is actually 3 clusters

8. Construct kmeans clusters

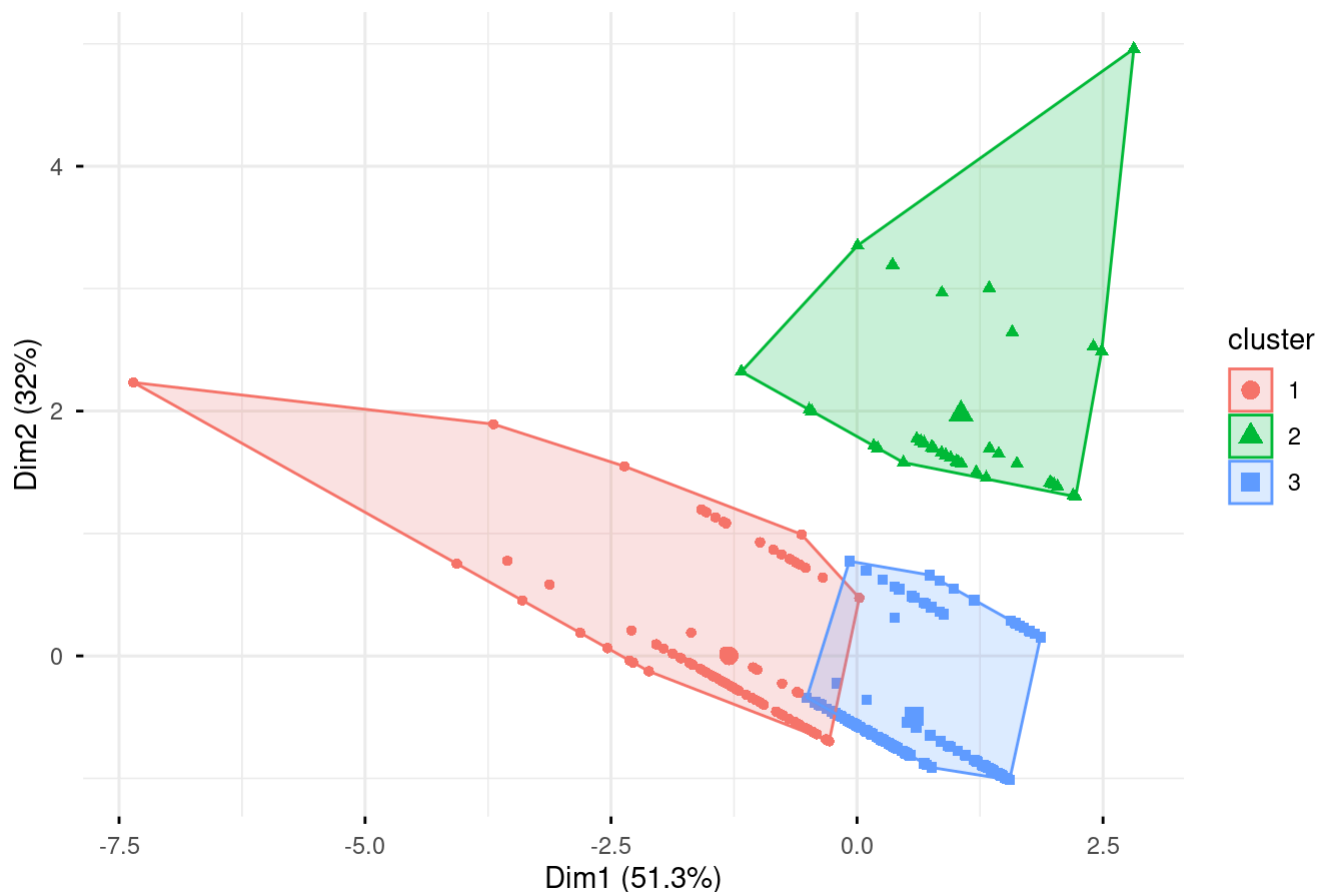
```
set.seed(420)
kmeans_3 <-
  kmeans(
    advantage_scaled,
    centers = 3 # create 3 clusters
  )
```

```
# Confirm clusters constructed
kmeans_3
```

9. Visualise clusters

```
fviz_cluster(
  kmeans_3,
  data = advantage_scaled,
  geom = "point",
  ggtheme = theme_minimal()
)
```

Cluster plot



Initial observation of optimal number of cluster might have suggested 5 is the optimal number of clusters. But with 3 clusters, the cluster plot displays the clusters with the least amount of overlaps.

This is important as we want unique and distinct clusters in order to observe any patterns that were not revealed via the rule-by-thumb segmentations. Having excessive clusters which aren't as distinct will likely prevent this.

10. Add clusters back to original

```
advantage_clusters <-  
  augment(kmeans_3, advantage_final)  
  
head(advantage_clusters)
```

A tibble: 6 × 11

	srch_id	prop_id	price_usd	hotel_star_ratings	children_booked	avg_availability
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1027	97708	158	3	0	0.333
2	1447	8729	329	5	0	0.5
3	2594	108493	120	2	0	-0.667
4	2615	74045	97	3	0	0.25
5	2654	17381	56	2	0	0.5
6	2740	89600	129	2	0	0.5

i 5 more variables: ptitle <int>, price_segment <chr>, rating_segment <chr>,
family <chr>, .cluster <fct>

11. Create final summary table (With clusters)

```

advantage_clust_summary <-
  advantage_clusters |>
  group_by(.cluster) |>
  summarise(
    bookings = n(),
    avg_price = mean(price_usd),
    avg_rating = mean(hotel_star_ratings),
    avg_children_booked = mean(children_booked),
    .groups = "drop"
  ) |>
  rename(Clusters = .cluster) |>
  arrange(desc(bookings))

advantage_clust_summary <-
  advantage_clust_summary |>
  kable(digits = 2)

advantage_clust_summary

```

Clusters	bookings	avg_price	avg_rating	avg_children_booked
3	160	103.93	2.64	0.19
1	104	188.67	4.00	0.18
2	40	114.80	2.83	2.28

General Analysis:

- 52.6% of bookings are from cluster 3, with the lowest average price and rating as well as children booked
- Majority of customers are not families, and books cheaper and lower rated hotels
- Cluster 1 could indicate the more premium customers which forms 28.3% of bookings but 46.3% of all spendings
- Cluster 2 is the family demographics which has the lowest number of bookings with 40, they might not be the most efficient group to focus on during competitive advantage contexts.

Part B. Analytics Team Memo (20 points)

Instructions

Length: 600 words. **Exhibits:** Maximum 2 figures or tables.

Write this as a bullet-point working document for your analytics peers and manager.

Bullets must be written as full sentences. Use Writing Tips from "Write Like an Amazonian". Only display code in memo if it is to define or explain. Do not produce any new analysis, table or figure in this section.

Memo

PLACE EXHIBITS IN APPROPRIATE SECTION AS NEEDED

1. Research Question

Research Question

What is the relationship between the pricing of a hotel, the rating of a hotel and the family demographic of customers across bookings that are competitive and not in terms of availability?

Why does this question matter?

Helps identify which factors result in the most number of bookings for Expedia. This provides Expedia the opportunity to push certain conditions to customers, gaining a competitive edge over other online travel agencies.

2. Data & Approach (5 pts)

Dataset used, key variables, and methods

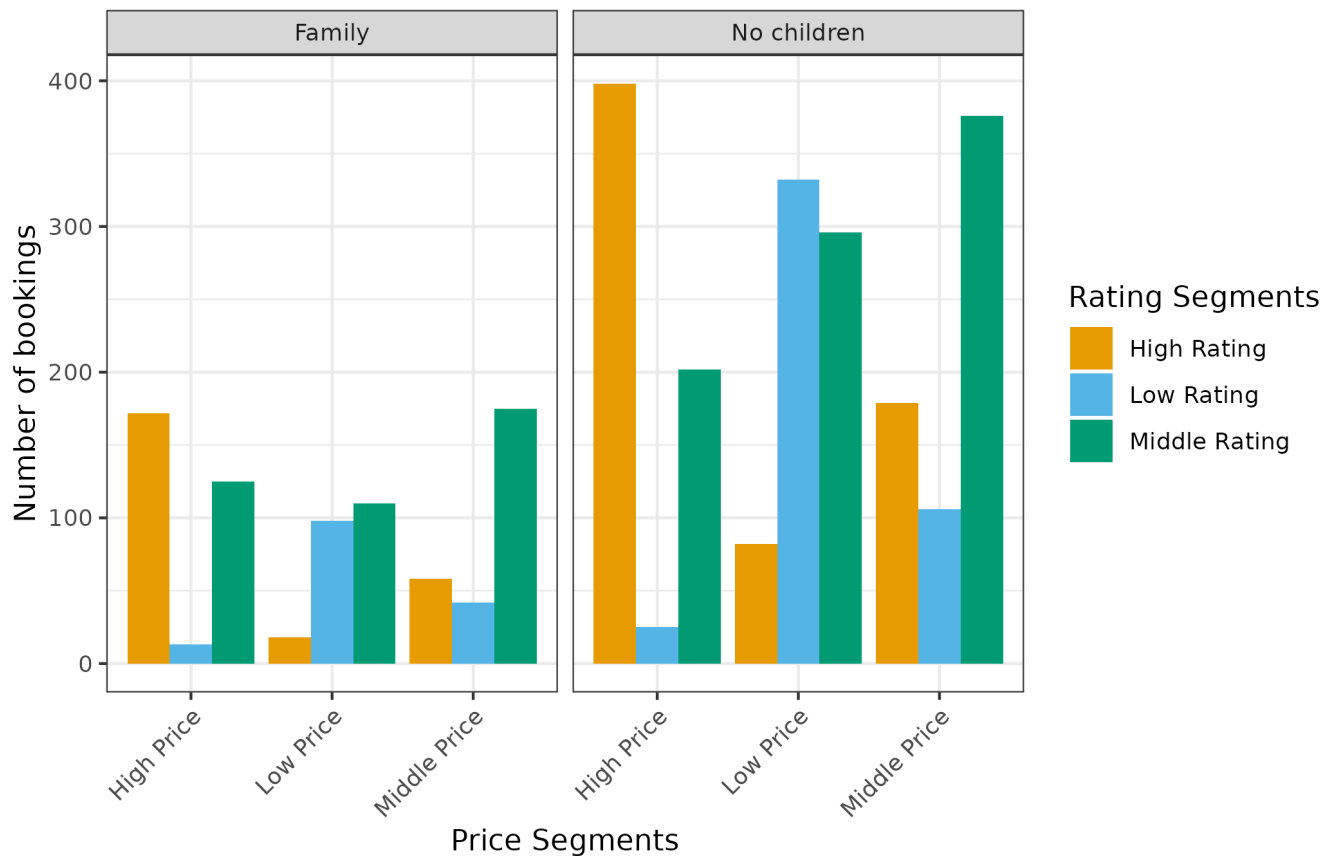
- 158269 observation data set was cut down to 3141 observations
- Cut down version was created by taking the variables: srch_id, prop_id, prop_starrating, gross_bookings_usd, price_usd, srch_children_count and all 8 comp_inv values. Then, filtering them to remove any missing values and to attain confirmed bookings (gross_bookings_usd > 0).
- Remaining comp_inv were compiled into a mean to identify average availability of competitors.
- Availability is regarded as competitive if the avg_availability does not equal 0 and is not when avg_availability value equals 0.
- Price was then grouped using ntiles, star ratings were grouped into buckets of 2, 3 and 4+ and families were defined as children_booked >= 1
- Rule of thumb segmentation was used with the ntiles and case_when. Kmeans clustering was used to identify more specific groupings with a cluster of 3 being chosen

3. Preliminary Findings (6 pts)

Non-competitive

Number of bookings by pricing, rating and family factor

For no competitive advantage

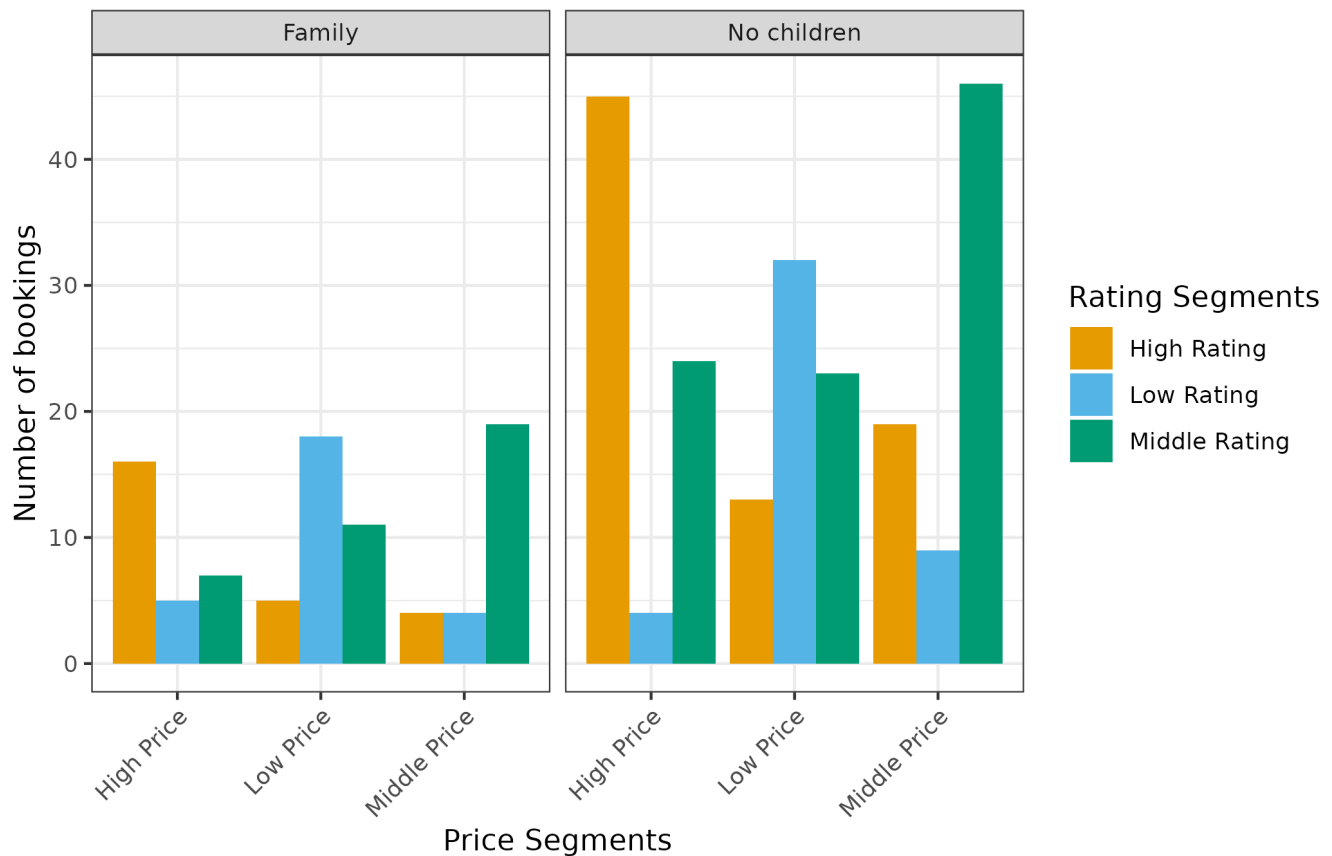


- 1996 and 812 bookings for non-family and family, respectively.
- Most popular conditions were high price and high rating (398 vs 172 family), followed by middle price and middle rating, suggesting a positive linear relationship between price and rating.
- Families book more high price, middle rating hotels (15.3% vs 10.1% non-family). Indicates families may be inclined to spend more on less premium hotels.
- Non-families favoured lower priced, lower rating hotels (17.3% vs 13.1% family). Suggested non-family are willing to settle for lower ratings at a lower price.

Competitive

Number of bookings by pricing, rating and family factor

For any competitive advantage



- Competitive advantage had modal conditions of middle price, middle rating (67 non-family and 26 family). Followed by high price, high rating (59 non-family and 26 family).
- This varies slightly from data of the "no competitive advantage" group, suggesting Expedia is better at selling high priced, high rated hotels than competitors when availability isn't limited.
- Middle priced and middle rated bookings are still the highest, which could suggest customers don't splurge, despite limited availability

Comparison

- A star rating above "low" is always preferred by customers. Only 21.9% of non-competitive and 24.3% of competitive bookings were for low rating hotels

Kmeans

- Kmeans clustering helps reveal specific groups that Expedia can focus to increase sales and bookings.
1. Budget non-family (55% of bookings, 42% of spendings) with an average spending of \$95 per booking
 2. Premium non-family (30.7% of bookings, 44.3% of spendings) with an average spending of \$179 per booking
 3. Family (14.3% of bookings, 13.7% of spendings) with an average spending of \$119 per booking

- Similar clusters for competitive advantage means availability has a minimal impact on customer behaviour.

4. Assumptions & Limitations (3 pts)

- Going from 158269 to 3141 observations could limit the reliability of the results and patterns.
- 304 bookings were situations where expedia had a competitive advantage which could produce inaccurate analysis.
- Families were assumed to be `children_booked >= 1` which might not address all instances defining a "family"
- Confirmed bookings were classified as `gross_bookings_usd > 0` and assumed to not be cancelled after payment.

5. Open Questions for Feedback (3 pts)

- Is it financially feasible for Expedia to implement different marketing strategies
- Should Expedia's algorithm deprioritise low rated hotels (less than 2 stars)

6. Next Steps (3 pts)

- Replace family demographic with more relevant variables like promotions
- Add price competitiveness to the analysis to investigate if that is a factor in Expedia bookings
- Investigate the booking details of bookings that fall into the premium non- family cluster to determine whether spending can be increased

Part C. Corporate Brief (20 points)

Instructions

Length: 400 words. **Exhibit:** 1 table OR figure.

Write this as an executive brief to be read and understood by managers.

Bullets must be written as full sentences. Use Writing Tips from "Write Like an Amazonian". No visible code in the brief. Do not produce any new analysis, table or figure in this section.

Executive Brief

Clusters	bookings	avg_price	avg_rating	avg_children_booked
1	1544	95.26	2.67	0.16
3	863	179.26	4.00	0.20
2	400	119.42	3.05	2.30

Executive Summary (5 pts)

The relationship between Expedia's hotel bookings was analysed based on price tier, star rating, family or non-family booking and level of competitor availability . The analysis was performed to assist Expedia in gaining a competitive edge over other online travel agencies. Across both levels of competition, customers consistently booked hotels that had similar price and rating levels. K means clustering also revealed a premium non-family customer segment that Expedia could take advantage of to improve their revenue generation.

Key Insights (5 pts)

- Despite contributing 24.3 percentage points fewer bookings than the budget non-family cluster, the premium non-family cluster generated \$84 more per booking on average. This places this group as the greatest value segment to target for revenue growth.
- Across both levels of competition, premium (high price, high rating) and standard (middle price, middle rating) hotels were the most booked . The former recorded 570 bookings (no advantage) and 85 (competitive advantage) while the latter recorded 537 bookings (no advantage) and 93 (competitive advantage), indicating that customers treat price and rating as a package rather than favouring one factor over the other.
- Families favoured high price/middle rating hotels 5.2 percentage points more than non families, while non families favoured low price/low rating 4.2 percentage points more than families. Low rated hotels contributed to the lowest, 21.9% and 24.3% respectively, and this volume declined sharply as prices rose, indicating star rating acts as a minimum quality filter before price is taken into account.

Business Implications (5 pts)

- The premium non-family segment generates \$84 more on average per booking than the budget non-family sector, indicating the retention and growth of this segment may be the most efficient to generate greater profits, especially at 3 star and above rated hotels where 77.7% of all bookings occur.
- Family and non-family bookers have different preferences in both price and rating, suggesting that promotional strategies may have to be increasingly tailored to be effective.
- The consistently low booking volume for low rated hotels suggest an opportunity to shift marketing resources towards middle and high rating hotels.

Recommended Actions (5 pts)

- Prioritise retaining and growing the premium non-family sector through promotions and retention schemes like loyalty points and discounts for customers who book these hotels.
- Trial promoting a greater proportion of middle rating, high price hotels to accounts that have previously had family bookings and then recompare this to the initial results to identify a boost in the sales of hotels to families.

References

- Schuckert, M., Liang, S., Law, R., & Sun, W. (2019). How do domestic and international high-end hotel brands receive and manage customer feedback? *International Journal of Hospitality Management*, 77, 528-537. <https://doi.org/10.1016/j.ijhm.2018.08.017>

Code Assistance

- Segmentation method was adapted from suggestions by Claude (Anthropic, 2026)
- Plot construction (angling labels) was adapted from suggestions by Claude (Anthropic, 2026)

Review:

- Some passages in the Analytics Team Memo were reviewed with the assistance of Claude (Anthropic, 2026), reviewed and edited by the authors.

Instructions:

- List any sources, readings, or external material you cited.
- Use a consistent referencing style (APA).

GenAI Use Declaration

How to Reference GenAI Use

If you used GenAI (e.g., ChatGPT, Copilot) for any part of your work, you must acknowledge it in **References** or a footnote. Examples:

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"Code snippets for data cleaning were adapted from suggestions generated by ChatGPT (OpenAI, 2025)."
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- Any use of GenAI must be **substantially adapted, verified, and integrated** into our group's own work.

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